

A Synthetic Multi-Band Antenna Design for Testing

Abstract - This paper presents a synthetic antenna design created for validation of automated figure extraction pipelines. The antenna operates at 2.4 GHz and 5.8 GHz with satisfactory return loss. Results are shown in Fig. 1 through Fig. 3.

I. INTRODUCTION Antenna design is a critical aspect of modern wireless communication systems. The proposed antenna in this work uses a rectangular patch on FR-4 substrate. The geometry of the antenna is shown in Figure 1. The substrate has a dielectric constant of 4.4 and thickness of 1.6mm. Optimizations were performed using HFSS to achieve dual-band operation.

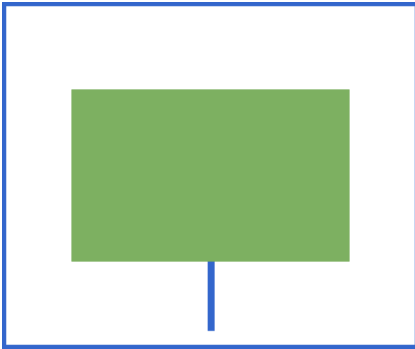


Fig. 1. Geometry of the proposed dual-band rectangular patch antenna.

The antenna was fabricated and measured using a vector network analyzer. The simulated and measured S11 parameters are compared in Fig. 2 and show good agreement. The radiation patterns shown in Fig. 3 confirm omnidirectional behavior in the H-plane.

The current distribution at 2.4 GHz shows concentration along the patch edges, confirming the fundamental TM₁₀ mode excitation. Further parametric studies on the feed position are discussed next.

II. ANTENNA DESIGN The proposed design employs a coplanar waveguide feed for improved impedance matching. The feed dimensions were optimized to achieve a VSWR below 2 across both operating bands. Table I summarizes the final dimensions of the antenna prototype.

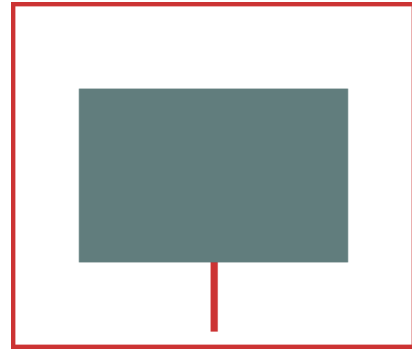


Fig. 2. Simulated and measured S11 parameters of the antenna.

III. RESULTS AND DISCUSSION The measured return loss at 2.4 GHz is -18.5 dB and at 5.8 GHz is -22.3 dB, which are well below the -10 dB threshold. The bandwidth at each resonance exceeds 100 MHz. The gain was measured to be 4.2 dBi at 2.4 GHz.



Fig. 3. Measured radiation patterns at 2.4 GHz and 5.8 GHz.

IV. CONCLUSION A dual-band rectangular patch antenna has been designed, fabricated, and tested. The antenna operates at 2.4 GHz and 5.8 GHz with return loss better than -15 dB at both bands as demonstrated in Fig. 2. The radiation patterns in Figure 3 show stable omnidirectional coverage. Future work will explore MIMO configurations.

REFERENCES [1] C. Balanis, *Antenna Theory*, 4th ed. Wiley, 2016. [2] R. Garg et al., *Microstrip Antenna Design Handbook*, Artech, 2001. [3] K. Wong, *Compact and Broadband Microstrip Antennas*. Fig. 5 in Ch. 3 provides a useful reference geometry for similar designs.